A REVIEW INTRODUCTION OF DATABASE MANAGEMENT SYSTEM

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Abstract - *Database* management system (DBMS) is computer software designed to manage the database is efficiently and conveniently. Database is structured to facilitate the storage, retrieval, modification and deletion of data in conjunction with various data processing operations. The main purpose of DBMS is to store and retrieve the data in the convenient and efficient way. This paper presents the overview of the database management system (DBMS) such as its component, Data Model which describe how data is to be arrange for a specific purpose. The major purpose of a database system is to provide users with an **abstract** view of the data .That are the system hides certain details of how the data are stored and maintained. The essential part of the database management system is security. It allows organizations to place control of database development in the hand of **database administrators (DBA)** and other specialist. It allows different user application programs to easily access the same database. A DBMS also provides the ability to logically present database information to users. With the increase of data to be indexed and retrieved and the increasing heavy workloads, modern search engines, reliability, distribution and performance problems. This paper presents a new and simple way for integration and compares the performance of our system to the current implementation based on storing the full text index directly on the file system.
Keywords: DBMS, Components, DBA, Entities, verbosity (redundancy).

1 INTRODUCTION

A database management system (DBMS) refers to the technology for creating and managing database. DBMS is software tool to organize (Create, Retrieve, Update, Delete, maintain and access the database). A database is an integrated collection of logically – related records or files consolidated into a common pool that provides data for one or more multiple users. DBMS is also charge of access, security, storage and host of other functions for the database system. It does through a selection of computer programs.

1.1 Database

Database (DB) is a collection of inter – related data which are used to retrieve, insert and delete the data efficiently. Data can be used to organize in the various formats such as table, schema, views, and reports, etc. for example: - The College Database organizes the data about the admin, staff, students, faculty etc. This can be easily insert and delete the information using database.

Database Management System (DBMS):- DBMS is software which is used to manage the database. It is a collection of program that enables users to create and maintain a database. This is the use most commonly use database examples are DBMS, MYSQL, oracle, etc. DBMS is the collection of those programs that are use to manage the database. The management of database includes creation, insertion, updation, manipulation and deletion of data. Database is secure and protected software.

2 APPLICATIONS OF DBMS:-

- In railway reservation system database is required to keep the record of ticket booking, train's departure and arrival status. It also required in the case of trains get late then people get to known it through database they get the update .
- Library management system is become easy in the library to track each book and maintain it because of the database. This happens because there are thousands of books in the library. It is very difficult to keep a record of all books in a copy or register. Now DBMS used to maintain all the information related to book issue dates, name of the book, author and availability of the book.
- Banking is one of the main applications of databases. We all known there will be a thousand transactions through banks daily and we are doing this without going to

the bank. This is all possible just because of DBMS that manages all the bank transactions.

- Human resource management keeps record of each employee's salary, tax and work through DBMS.
- In universities and colleges Examinations are done online basis and universities as well as colleges maintain all these records through DBMS such as student 's registrations details, results, courses and grades all the information are stored in database.
- In military to keeps the records of millions of soldiers and it have millions of files keeping secured and safe. As DBMS provides a big security assurance to the military. It is widely used in militaries. Someone can easily search for all the information in seconds with the help of DBMS.

2.1 Components of DBMS

The major components of DBMS are as follows.

1). Hardware: -Hardware includes the actual physical computer system that is used to access the database. Mainframes to microcomputers are utilized in DBMS. Oftentimes large storage devices are used to store a huge amount of data.

2). Software: - The DBMS is the software here. It helps connect the physical database to the end users whenever a user wants to access the database; access has to be granted by the software.

3). Interface drivers: - A user or application program initiates either schema modification or content modification . These drivers are built on top of SQL. They provide methods to prepare statements execute statements, fetch results, etc. Examples include DDL, DCL, DML, etc.

4).Database Engine:- The database engine manages the core services of storing, processing, and securing data. It typically provides controlled access and rapid transaction processing to meet the needs of the most demanding data-consuming applications.

The database engine is most commonly integrated when creating relational databases for online transaction processing or online analytical processing of data.

5). Storage Engine: -This component stores and retrieves data records. It also provides a mechanism to store metadata and control information such as undo logs, redo logs, lock tables, etc.

6). Query Language :- DQL is a portion of SQL statements that allows you to get and organize data from a database. You can use the SELECT command to extract data from a database in order to perform action on it. It is same as the projection operation in relational algebra.

3 DATABASE ADMINISTRATOR (DBA)

A database Administrator uses as specialized software to store and organize data. The role may include capacity planning, installation, configuration, database design, migration, performanance, monitoring, security, troubleshooting, as well as backup and data recovery. Database administrator is a major key function in any firm or organization that is relying on one or more databases. They are overall commanders of the database system.

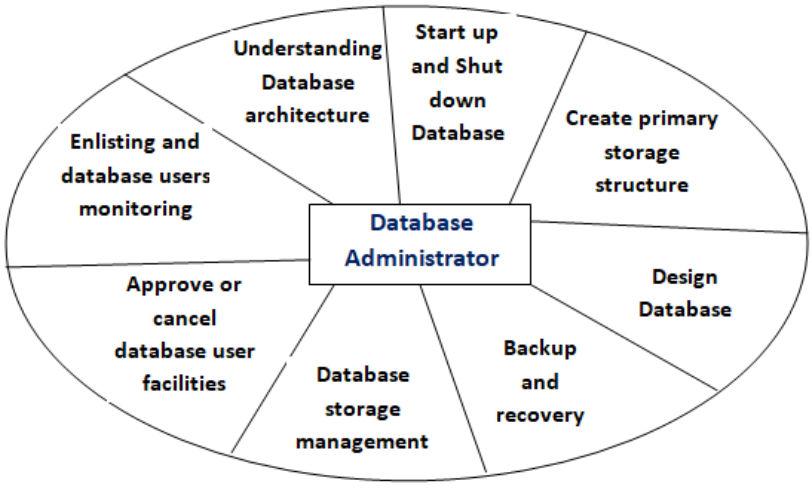
Tasks	Skills	Roles
•Developing, administering and maintaining databases. •Understanding the requirements for data stroage and accessibility.	•Good understanding of networks, data protection and databases. •Familiarity with databases such as Oracle and MySQL.	•Database design •Database accessibility •Performance issues •Capacity issues •Data replication •Table maintenance

4 IMPORTANCE OF DATABASE ADMINISTRATOR (DBA)

- Database Administrator manages and controls three level of database which is internal level, conceptual level and external level of database management system (DBMS) architecture and in discussion with the comprehensive user community, give a definition of the world view of database. Then it provides external view of different users and application.
- It grants permission to users of the database and contains a profile of each and every user in the database.
- Database administrator is only responsible for reducing the risk of data loss as it backup the data at regular intervals.
- It's responsible for performing regular system backups no data is lost during the power outage or other type of disaster.

4.1 Role of database administrator (DBA)

- **Decides hardware:** -They decide on economical hardware, based on cost, performance, and efficiency of hardware and best suits the organization. It is hardware that is an interface between end users and the database.
- **Manage data integrity and security: -** Data integrity needs to be checked and managed accurately as it protects and restricts data from unauthorized use. DBA eyes on relationships within data to maintain data integrity.
- **Database design:-** DBA is held responsible and accountable for logical, physical design, external model design, and integrity and security control.
- **Database implementation:-** DBA implements DBMS and checks database loading at the time of its implementation.
- **Query processing performance:-** DBA enhance query processing by improving speed, performance, and accuracy.

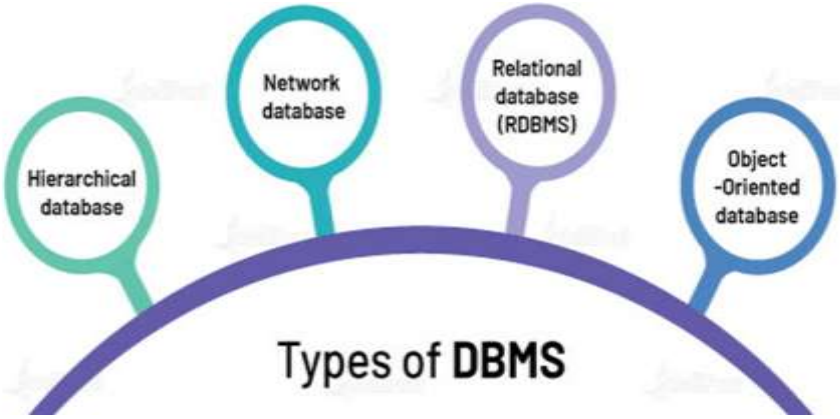


5 VARIOUS RESPONSIBILITIES OF DATABASE ADMINISTRATOR (DBA)

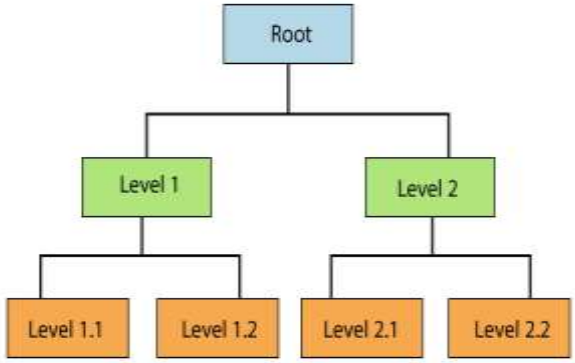
- Responsible for designing overall database schema (tables & fields).
- To select and install database software and hardware.
- Responsible for deciding on access methods and data storage.
- DBA selects appropriate DBMS software like Oracle, SQL server or MySQL.
- Used in designing recovery procedures.
- DBA is responsible for specifying various techniques for monitoring the database performance.

5.1 Types of DBMS:-

There are various types of database used for storing different varieties of data.



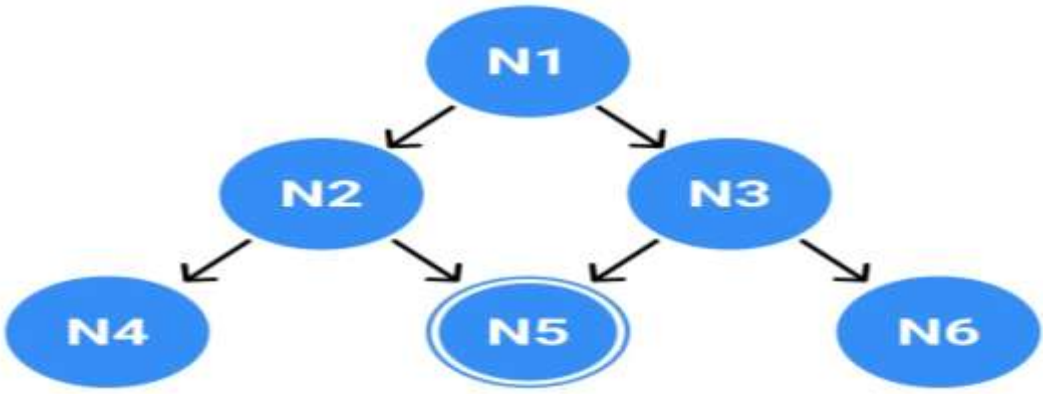
- **Hierarchical Database :-** This is one of the oldest model in a data model which was developed by IBM, in the 1950s. In a hierarchical model, data are viewed as a collection of tables, or we can say segments that form a hierarchical relation . In this , data is organized into a tree-like structure where each record consists of one parent record and many children .



Hierarchical Database

Data get stored in the form of records that are connected via links. Each child record in the tree will contain only one parent. On the other hand , each parent record can have multiple child records.

- **Network Database : -** It is database that typically follows the network data model . Here, the representation of data is in the form of nodes connected via links between them. Unlike the Hierarchical database, it allows each record to have multiple children and parent nodes to from a generalized graph structure.



- **Relational Database (RDBMS) :-** This database is based on the relational data model, which stores data in the form of rows(tuple) and columns(attribute) , and together forms a table(relation). A relational database uses SQL for storing , manipulating , as well as maintaining the data . Each data in the database carries a key that makes the data unique from others. **Example** of relational database is MySQL , Microsoft SQL server , Oracle etc.

5.2 Properties of Relational Database:

Relational database have the following properties.

- Values are atomic.
- All of the values of column have the same data type.
- Each row is unique.
- The sequence of columns is insignificant.
- The sequence of rows is insignificant.
- Each column has aunique name.
- Integrity constraints maintain data consistency across multiple tables.

- **Object – oriented Database :-** The type of database that uses the object – based data model approach for storing data in the database system . The data is represented and stored as objects used in the object – oriented programming language.

5.3 Components of Object Oriented Database:

1). Objects :- An object is an abstraction of a real world entity or we can say it is an instances of class . Objects encapsulates data and code into a single unit which provide data abstraction by hiding the implementation details from the user .

2). Attribute :- An attribute describes the properties of object . For example : object is STUDENT and its attribute are Roll no, Branch, seamarks() in the student class.

3). Methods :- Method represents the behavior of an object . Basically,it represents the real world action.

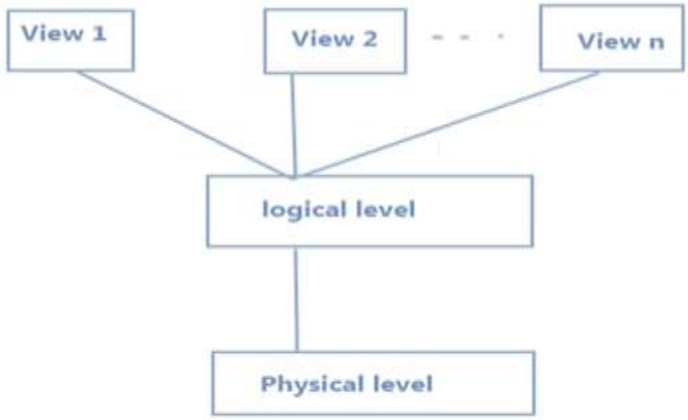
4). Class :- A class is a collection of similar objects with shared structure i.e. attributes and behavior i.e. methods . An object is an instance of class.

6 DATA ABSTRACTION

Data Abstraction is a process of hiding unwanted or irrelevant details from the end user . it provides a different view and helps in achieving data independence which is used to enhance the security of data .

Mainly three levels of data abstraction are possible in DBMS.

- Physical or Internal Level
- Logical or conceptual Level
- View or External Level



Three Levels of data abstraction

6.1 Physical or Internal Level

It is the lowest level of abstraction for DBMS which defines how the data is actually stored. The physical level describes complex low – level data structures in detail. It defines data – structure to store data and access methods used by the database. Actually, it is decided by developers or database application programmers how to store the data in the database.

6.2 Logical or Conceptual Level

The next – higher level of abstraction describes what data are stored in the database, and what relationship exist among those data. The logical level thus describes the entire database in terms of small number of relatively simple structures. Database administrators, who must decide what information to keep in the database, use the logical level of abstraction.

View or External Level

The highest level of abstraction describes only part of the entire database. Even through the logical level uses simpler structures, complexity remains because of the variety of information stored in a large database. Many users of the database system do not need all this information, instead, they need to access only a part of the database. The view level of abstraction exists to simplify their interaction with the system. The system may provide many views for the same database.

7 DATABASE LANGUAGE

DBMS has appropriate language and interfaces through with user can express queries and updates. To read, store, insert the data in the database DBMS language are used.

A database query language and report object allows users to interactively interrogate the database, analyze its data, and update it according to the users privileges on data. It also controls the security of the database. Data security prevents unauthorized users from viewing or updating the database. Using passwords, users are allowed a access to the entire database or subsets of it called subschema.

8 CONCLUSION

Like other software database management system also have some advantages or disadvantages. DBMS can improve data processes and increases the business value of your organization. Database software is designed for update large volume of data. DBMS remove the data redundancy, duplication and manage the consistency. Userstake help of DBMS on managing the huge amount of data. Data is stored in the form of record. For inserting, deleting, modifying records data manipulation language (DML) is used. Data model are necessary for introducing abstraction. Data model specify that how data link to one another, as well as handled and stored in the system. Either programmer or one who implement the database will have much more work to do without using DBMS. Operations such as to arrange data or inserting data into the database and also obtaining out from the database it will become difficult. The Database Administrator (DBA) performance the some tasks are Developing and maintaining databases and also understanding the requirements for data storage and accessibility. A database administrator has responsibilities of monitoring the database and the movement of data in the database. The major role of data administrator is design the database and also checks the backup and recovery of data in database.

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